

Batch: A3 Roll No.: 16010324044
Experiment / assignment / tutorial No. 3
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Clamper circuit.

AIM: To analyse given clamper circuit. To design clamper circuit for given transfer characteristics.

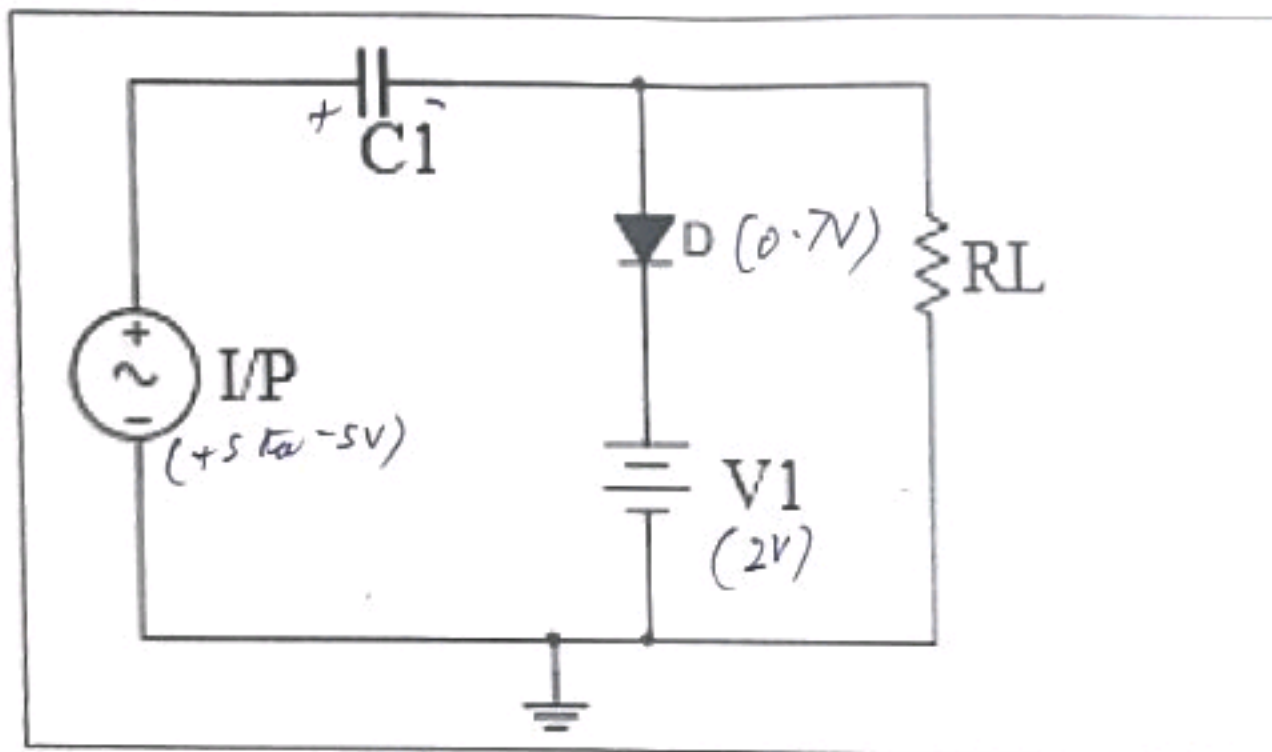
1. **OUTCOME:** Student will be able to analyze and design diode circuits.

Part I- Analysis of clamper circuit

Procedure:

1. Develop the circuit in hardware as shown in the diagram.
2. Apply sinusoidal waveform 10V(P-P) as input and DC voltage (V1) at the given value of 2 V. Select $C1 = 0.1 \mu F$, $R_L = 100 K$.
3. Observe the waveforms.
4. Identify the type of Clamper circuit.
5. Increase the DC voltage (V1) to 4V, and notice to what voltage are the peaks clamped off.
6. If the DC source is reversed, how does this affect the clamping?

Circuit Diagram:



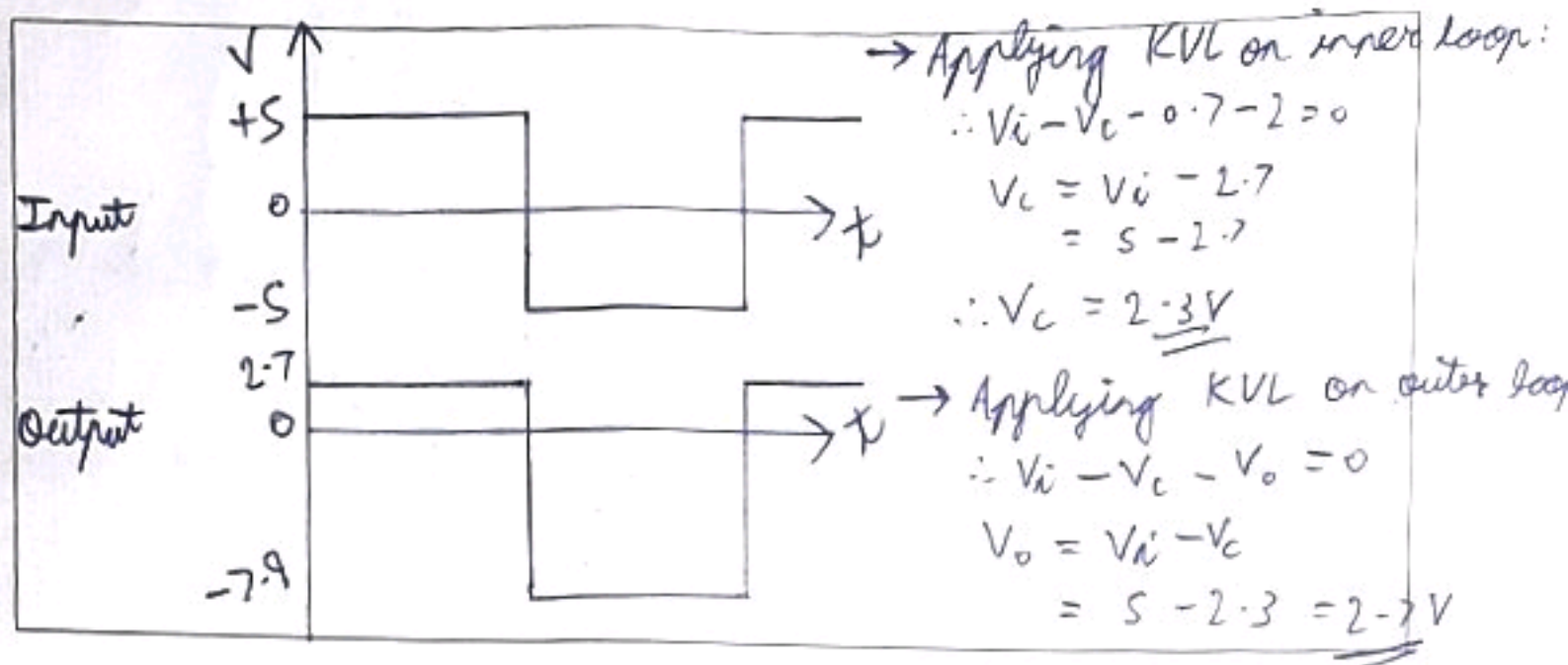
Observation Table I:

	Vin (max)	Vin (min)	Vout (max)	Vout (min)

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Theoretical	5V	-5V	2.7V	-7.3V
Practical	5.1V	-4.9V	3.3V	-8.1V

Input and output waveforms

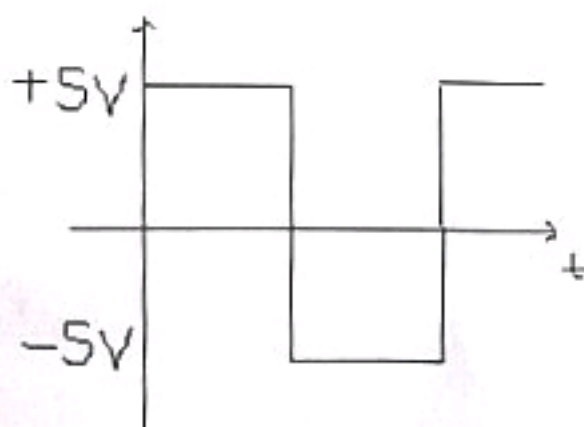


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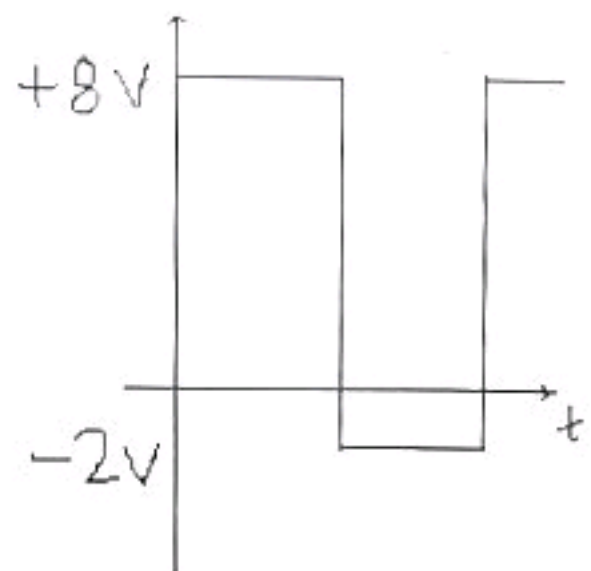
Part II- Design of Clamper circuit

1. Design a clamper circuit to perform the function indicated in the following diagram

Input



Output



Procedure:

1. Set the clipping levels (biasing voltage) as per the voltage transfer function.
2. Implement the designed circuit.

3. Plot input, output waveforms.

Design Steps:

- The first diagram represents a negative clamper.
 - To convert it into a positive clamper, we must change the position of diode; which eventually changes the orientation of capacitor too.
 - Now, applying KVL on inner loop:

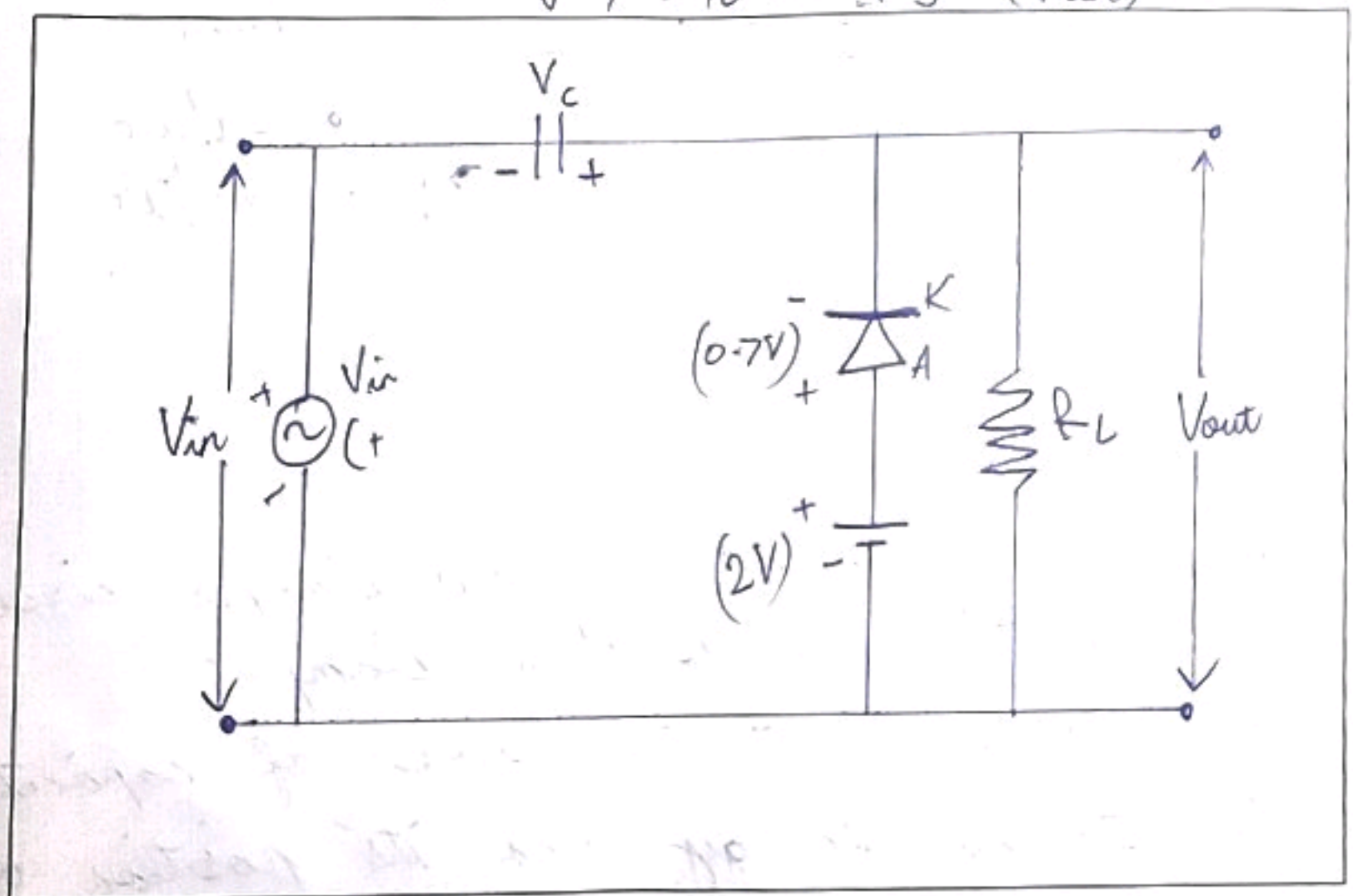
$$\therefore V_i + V_c + 0.7 - 2 = 0$$

$$\therefore 5 + V_c + 0.7 - 2 = 0$$

$$\therefore V_c = 3.7V$$
 - Now, applying KVL on outer loop:

$$\therefore V_i + V_c - V_o = 0$$

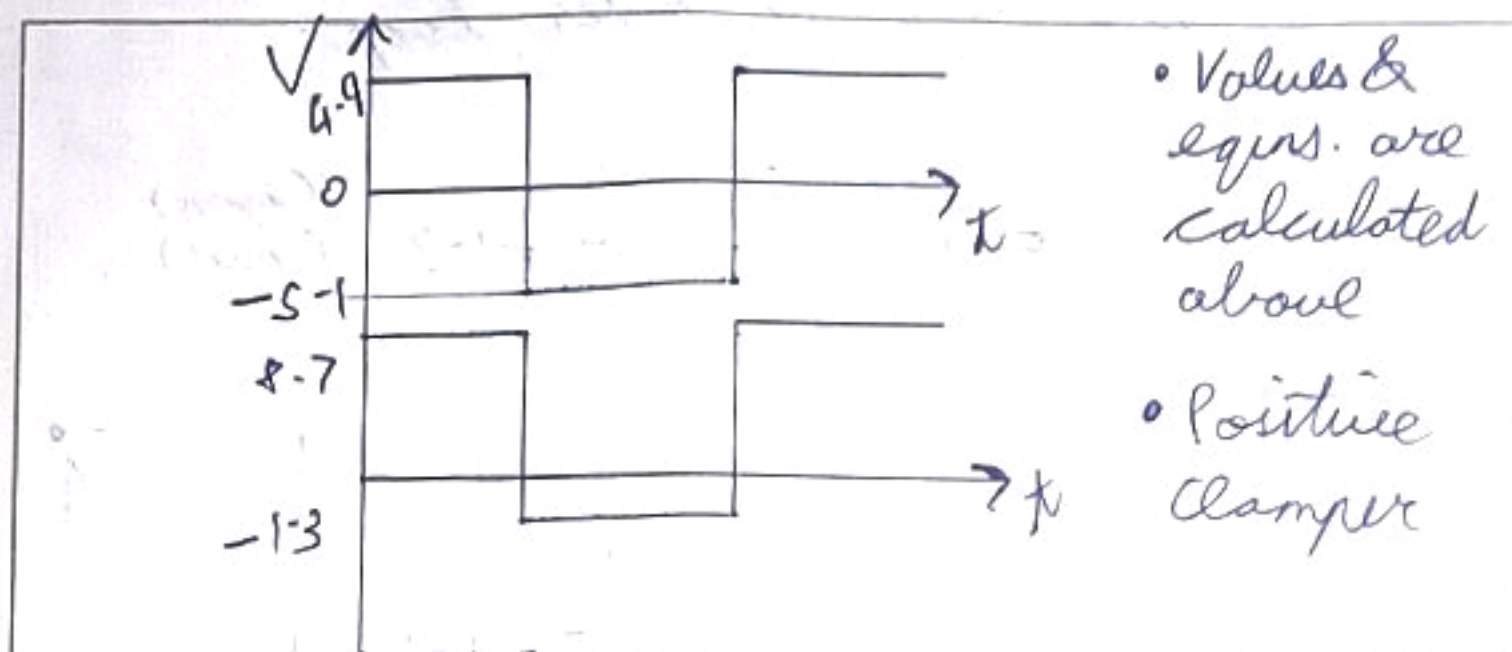
$$\therefore V_o = V_i + V_c = 5 + 3.7 = 8.7V \quad (\text{max})$$
- Circuit Diagram: $5 + 8.7 - 10 = -1.3 \quad (\text{min})$



Observation Table II:

	Vin (max)	Vin (min)	Vout (max)	Vout (min)
Theoretical	5V	-5V	8V	-2V
Practical	4.9V	-5.1V	8.7V	-1.3V

Input and output waveforms



Conclusion:

- We learnt about various clamper circuit i.e. positive & negative clamper
- We also learnt the role of capacitor in storing energy and its position and orientation in various clamper circuit.

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